Chapter 5: Differentiation

Calculus II

March 24, 2025

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Section 1

Differentiable functions

Differentiability of Single-Variable Functions

Recall:

For a function f of a single variable x, we say that f is **differentiable at** a if f is **locally linear** at a.

That is, there exists a constant m such that for all h sufficiently close to zero,

$$f(a+h) - f(a)$$

is **well approximated** by mh.

By "well approximated", we mean that the difference between $f(a+h) - f(a)$ and mh is small compared to h when h is small:

$$\frac{(f(a+h) - f(a)) - mh}{h} \to 0 \quad \text{as} \quad h \to 0.$$

In this case, we say f is differentiable at a, and we write

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = m.$$

Derivative and Linear Approximation

Derivative Definition:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = m.$$

Key Idea

The number m is called the derivative of f at a, denoted $f'(a)$.

Linear Approximation:

$$f(x) \approx f(a) + f'(a)(x - a).$$

Key Idea

This is the best linear approximation of f at a.

Extending to Functions from $\mathbb{R}^2$ to $\mathbb{R}$

Now let's extend the notion of "local linearity" to functions from $\mathbb{R}^2$ to $\mathbb{R}$.

- Recall from linear algebra that a linear function ℓ from $\mathbb{R}^2$ to $\mathbb{R}$ is of the form

$$\ell(h, k) = ph + qk, \tag{5.1}$$

  where p and q are constants.

- Recall the **norm** of a 2-dimensional vector:

$$\|(h, k)\| = \sqrt{h^2 + k^2}.$$

Definition 5.1.1

A function f defined in an open disk (in $\mathbb{R}^2$ we say open disk instead of open ball) in $\mathbb{R}^2$ centered at (a, b) is differentiable at (a, b) if

$$f(a + h, b + k) - f(a, b)$$

can be well approximated by a linear function ℓ in the following sense:

$$\frac{(f(a + h, b + k) - f(a, b)) - \ell(h, k)}{\|(h, k)\|} \tag{5.2}$$

tends to zero as $\|(h, k)\|$ tends to zero.

We call

$$L(x, y) = f(a, b) + \ell(x - a, y - b)$$

the linear approximation of $f(x, y)$ at (a, b).

Vector notation version

Definition 5.1.1 can be rewritten in vector language:

Let

$$\mathbf{A} = (a, b) \quad \text{and} \quad \mathbf{H} = (h, k)$$

Definition 5.1.2 (Vector notation version)

A function f from $\mathbb{R}^2$ to $\mathbb{R}$ defined on an open disk centered at $\mathbf{A}$ is differentiable at $\mathbf{A}$ if

$$f(\mathbf{A} + \mathbf{H}) - f(\mathbf{A})$$

can be well approximated by a linear function ℓ in the following sense:

$$\frac{(f(\mathbf{A} + \mathbf{H}) - f(\mathbf{A})) - \ell(\mathbf{H})}{\|\mathbf{H}\|}$$

tends to zero as $\|\mathbf{H}\|$ tends to zero.

Theorem 5.1.3

If a function f from $\mathbb{R}^2$ to $\mathbb{R}$ is differentiable at $\mathbf{A}$ then f is continuous at $\mathbf{A}$.

Proof:

By Definition 5.1.2,

$$f(\mathbf{A}+\mathbf{H}) - f(\mathbf{A}) - \ell(\mathbf{H}) \to 0 \quad \text{as} \quad \|\mathbf{H}\| \to 0.$$

Since $\ell(\mathbf{H}) \to 0$ as $\|\mathbf{H}\| \to 0$, it follows that

$$f(\mathbf{A}+\mathbf{H}) - f(\mathbf{A}) \to 0.$$

Therefore, f is continuous at $\mathbf{A}$.

Relating to Partial Derivatives

Next we relate the numbers p and q in the linear function $\ell(h,k) = ph + qk$ in formula (5.2) to the function f.

- Suppose f is differentiable at (a,b) and $\ell(h,k) = ph + qk$.
- Set $k = 0$ in Definition 5.1.1; then

$$\lim_{h\to 0} \frac{f(a+h,b) - f(a,b) - ph}{h} = 0.$$

- This shows that $f(x,b)$ is differentiable in x at a, and

$$p = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}.$$

- The number p is called the **partial derivative of f with respect to x** at (a,b), denoted

$$\frac{\partial f}{\partial x}(a,b) \quad \text{or} \quad f_x(a,b).$$

Thus, $\frac{\partial f}{\partial x}(a,b)$ is found by holding y equal to the constant b and differentiating $f(x,b)$ with respect to x at a.

Continue

Similarly, set $h = 0$ in Definition 5.1.1; then

$$\lim_{k \to 0} \frac{f(a, b+k) - f(a, b) - qk}{k} = 0.$$

- This shows that $f(a, y)$ is differentiable in y at b, and

$$q = \lim_{k \to 0} \frac{f(a, b+k) - f(a, b)}{k}.$$

- The number q is called the **partial derivative of f with respect to y** at (a, b), denoted

$$\frac{\partial f}{\partial y}(a, b) \quad \text{or} \quad f_y(a, b).$$

Continue

If f is differentiable at (a, b), then:

- It has partial derivatives $f_x(a, b)$ and $f_y(a, b)$.
- The linear approximation of f at (a, b) is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Rules for Partial Derivatives:

(Following from the rules for ordinary differentiation)

(a) $(f + g)_x = f_x + g_x$ and $(f + g)_y = f_y + g_y$

(b) $(fg)_x = f_x g + f g_x$ and $(fg)_y = f_y g + f g_y$

(c) $\left(\frac{1}{f}\right)_x = -\frac{f_x}{f^2}$ and $\left(\frac{1}{f}\right)_y = -\frac{f_y}{f^2}$

Example 5.1.4

Show that $f(x, y) = xy^2$ is differentiable at $(1, 3)$.

Proof:

Step 1: Compute Partial Derivatives

- Holding y fixed and differentiating f with respect to x we get:

$$f_x = y^2, \quad f_x(1, 3) = 9.$$

- Holding x fixed and differentiating f with respect to y we get:

$$f_y = 2xy, \quad f_y(1, 3) = 6.$$

Proof (Continued)

Step 2: Define Local Linear Approximation

- We have p and q and check that f is locally linear at $(1, 3)$, where

$$\ell(h, k) = 9h + 6k.$$

- Compute the error term:

$$\frac{f(1+h, 3+k) - f(1,3) - \ell(h,k)}{\|(h,k)\|}$$

$$= \frac{(1+h)\left(3^2 + 6k + k^2\right) - (1)(3)^2 - (9h + 6k)}{\|(h,k)\|}$$

$$= \frac{k^2 + 6hk + hk^2}{\|(h,k)\|}.$$

Proof (Continued)

Step 3: Bounding the error term

- Using the triangle inequality:

$$|k^2 + 6hk + hk^2| \leq k^2 + 6|hk| + |h|k^2.$$

- Since we are taking the limit as $\|(h,k)\| \to 0$ we can restrict attention to where $\|(h,k)\| \leq 1$, and there $|h| \leq 1$. We get:

$$\left|k^2 + 6hk + hk^2\right| \leq (1 + |h|)k^2 + 6|hk| \leq 2k^2 + 6|hk|. \tag{5.3}$$

- Further, using $(h \pm k)^2 = h^2 + k^2 \pm 2hk \geq 0$:

$$2|hk| \leq h^2 + k^2. \tag{5.4}$$

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Proof (Continued)

- Finally, with the inequalities (5.3) and (5.4), if $\|(h,k)\| \leq 1$ we have:

$$\frac{\left|k^2+6hk+hk^2\right|}{\|(h,k)\|} \leq \frac{2k^2+6|hk|}{\|(h,k)\|} \leq \frac{2k^2+3h^2+3k^2}{\|(h,k)\|} \leq \frac{5\left(h^2+k^2\right)}{\|(h,k)\|}$$

$$= 5\sqrt{h^2+k^2}$$

Therefore, since:

$$\frac{f(1+h,3+k)-f(1,3)-\ell(h,k)}{\|(h,k)\|} \to 0 \quad \text{as} \quad \|(h,k)\| \to 0,$$

we can conclude that $f(x,y)=xy^2$ is differentiable at $(1,3)$.

The next example shows that the existence of partial derivatives, while **necessary** for differentiability at a point, is **not sufficient**.

Example 5.1.5

Define a function f as

$$f(x,y) = |x+y| - |x-y|. \tag{5.5}$$

- The single variable functions are:

$$f(x,0) = |x+0| - |x-0| = 0, \quad f(0,y) = |0+y| - |0-y| = 0.$$

- Both are constant and differentiable at $(0,0)$, with partial derivatives:

$$f_x(0,0) = 0, \quad f_y(0,0) = 0.$$

Example 5.1.5 cont'd

- Next, we show that f is not differentiable at $(0,0)$ by evaluating:

$$\frac{f(0+h,0+k)-f(0,0)-\ell(h,k)}{\|(h,k)\|},$$

  where $\ell(h,k)=0h+0k$.

- Take $k=h$; then:

$$\frac{f(0+h,0+h)-f(0,0)-(0h+0h)}{\sqrt{h^2+h^2}}=\frac{|2h|-0-0}{\sqrt{2h^2}}=\frac{2}{\sqrt{2}}.$$

- This does not tend to zero as $\|(h,h)\|\to 0$. Therefore, f is not differentiable at $(0,0)$.

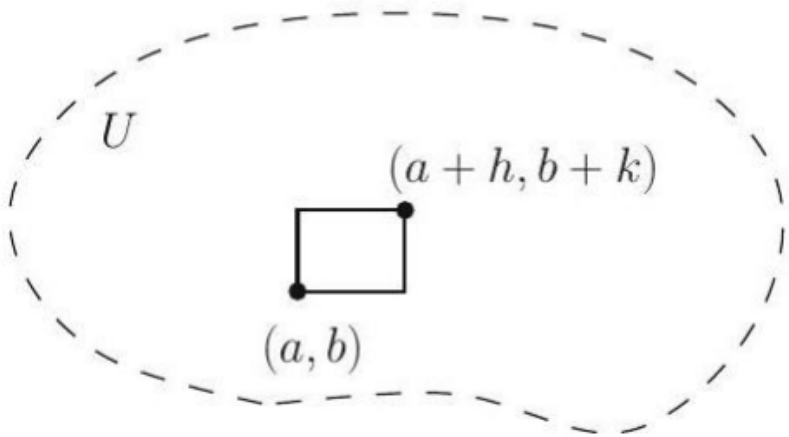
Theorem 5.1.6 (Mean Value Theorem)

Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$ whose partial derivatives f_x and f_y exist on an open set U containing (a, b).

Then for each (h, k) with $\|(h, k)\|$ sufficiently small, there exist numbers h' and k' such that:

$$f(a+h, b+k) - f(a, b) = hf_x(a+h', b+k) + kf_y(a, b+k'), \tag{5.6}$$

where $a + h'$ lies between a and $a + h$, and $b + k'$ lies between b and $b + k$.



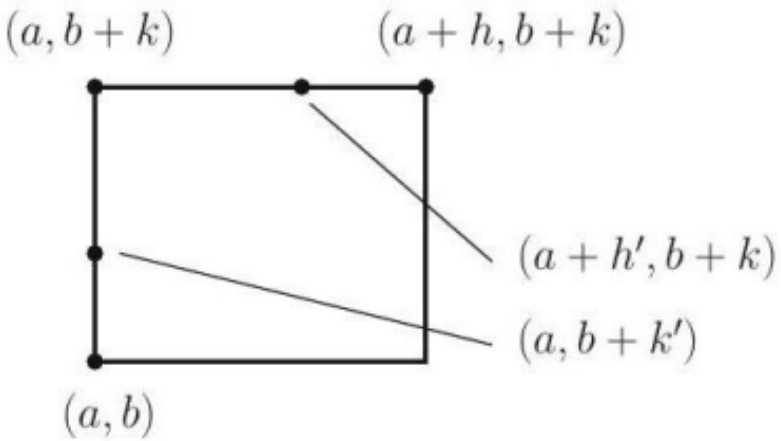
Proof

Write

$$f(a+h,b+k)-f(a,b)=f(a+h,b+k)-f(a,b+k)+f(a,b+k)-f(a,b)$$

- For $|h|$ and $|k|$ sufficiently small, all points on the sides of a small rectangle are in U, and $f(x,b+k)$ is differentiable on the interval from a to $a+h$.
- Apply the **Mean Value Theorem** for single-variable functions to $f(x,b+k)$: there exists $a+h'$ between a and $a+h$ such that

$$f(a+h,b+k)-f(a,b+k)=hf_x(a+h',b+k).$$

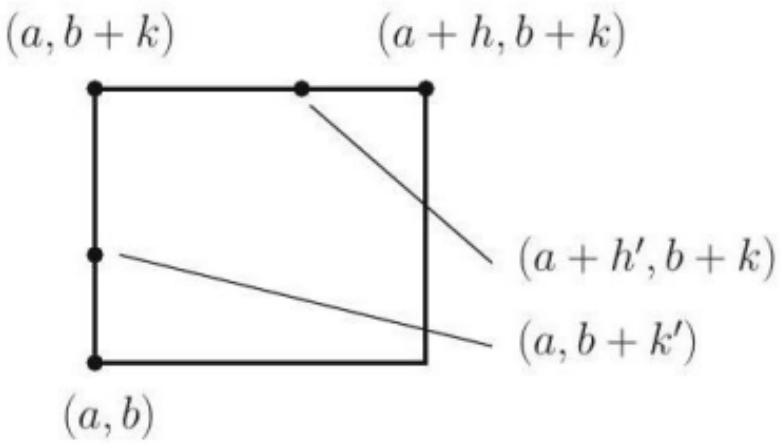


Proof (Continued)

- Similarly, $f(a, y)$ is differentiable on the interval between b and $b + k$, and there exists $b + k'$ between b and $b + k$ such that

$$f(a, b + k) - f(a, b) = k f_y(a, b + k').$$

- Add these two equations to obtain formula (5.6), completing the proof.



Theorem 5.1.7

If the partial derivatives of f are continuous in an open set containing (a, b), then f is differentiable at (a, b).

Proof.

Using formula (5.6) of **Theorem 5.1.7**, we have:

$$\frac{f(a+h, b+k) - f(a,b) - (hf_x(a,b) + kf_y(a,b))}{\sqrt{h^2+k^2}}$$

$$= \frac{hf_x(a+h', b+k) + kf_y(a, b+k')) - (hf_x(a,b) + kf_y(a,b))}{\sqrt{h^2+k^2}}$$

$$= \frac{h(f_x(a+h', b+k) - f_x(a,b)) + k(f_y(a, b+k') - f_y(a,b))}{\sqrt{h^2+k^2}} \tag{5.7}$$

Here, h' is between 0 and h and k' is between 0 and k.

Proof (Continued)

By the triangle inequality, the absolute value of the last expression (5.7) is less than or equal to

$$\frac{|h|}{\sqrt{h^2+k^2}}\left|f_x\left(a+h',b+k\right)-f_x(a,b)\right|+\frac{|k|}{\sqrt{h^2+k^2}}\left|f_y\left(a,b+k'\right)-f_y(a,b)\right|.$$

- Since $\frac{|h|}{\sqrt{h^2+k^2}}$ and $\frac{|k|}{\sqrt{h^2+k^2}}$ are less than or equal to 1, and f_x and f_y are continuous, each term in the last sum tends to zero as $\|(h,k)\| \to 0$.
- Therefore, (5.7) tends to zero as $\|(h,k)\| \to 0$, proving that f is differentiable at (a,b).

Hyperbolic functions

- Hyperbolic sine:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

- Hyperbolic cosine:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Their derivatives satisfy

$$\frac{d}{dx} \sinh x = \cosh x$$

$$\frac{d}{dx} \cosh x = \sinh x$$

Example 5.1.8

Let:

$$f(x, y) = y + \sin(xy) + \sinh x$$

Use the linear approximation of f at $(0, 1)$ to approximate $f(0.1, 0.9)$.

The linear approximation of f at $(0, 1)$ is

$$f(x,y) \approx L(x, y) = f(0, 1) + f_x(0, 1)(x - 0) + f_y(0, 1)(y - 1).$$

1 2

We calculate

$$f(0, 1) = 1 + \sin(0) + \sinh(0) = 1 + 0 + \tfrac{1}{2}\left(\mathrm{e}^0 - \mathrm{e}^{-0}\right) = 1$$

$$f_x(x, y) = \cos(xy)y + \cosh x, \quad f_x(0, 1) = \cos(0)1 + \tfrac{1}{2}\left(\mathrm{e}^0 + \mathrm{e}^{-0}\right) = 2$$

$$f_y(x, y) = 1 + \cos(xy)x, \qquad f_y(0, 1) = 1 + \cos(0)0 = 1.$$

Therefore $L(x, y) = 1 + 2(x - 0) + 1(y - 1)$ and

$$L(0.1, 0.9) = 1 + 2(0.1) + 1(0.9 - 1) = 1 + 0.2 - 0.1 = 1.1$$

is a good approximation of $f(0.1, 0.9) = 1.09005\ldots$.

Partial derivatives and differentiability for $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$

We extend the definitions of differentiability, local linearity, and partial derivative to functions $f : \mathbb{R}^n \to \mathbb{R}$.

- The partial derivative with respect to the i-th variable is defined while holding the others fixed:

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_n)}{h}.$$

- Let $\mathbf{F}$ be a function $\mathbb{R}^n \to \mathbb{R}^m$ whose component functions

$$\mathbf{F}(x_1, \ldots, x_n) = (f_1(x_1, \ldots, x_n), f_2(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_n))$$

  are differentiable.

- Each component function $f_i : \mathbb{R}^n \to \mathbb{R}$ has n partial derivatives, denoted $\frac{\partial f_i}{\partial x_j}$ or f_{i,x_j}.

Continue

The partial derivatives of $\mathbf{F}$ can be arranged into an $m \times n$ matrix, called the **matrix derivative** of $\mathbf{F}$ at $\mathbf{A}$:

$$D\mathbf{F}(\mathbf{A}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{A}) & \frac{\partial f_1}{\partial x_2}(\mathbf{A}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{A}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{A}) & \frac{\partial f_2}{\partial x_2}(\mathbf{A}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{A}) \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{A}) & \frac{\partial f_m}{\partial x_2}(\mathbf{A}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{A}) \end{bmatrix} \tag{5.8}$$

Using vectors and matrices we can express the definition of differentiability of $\mathbf{F}$ at $\mathbf{A}$.

Definition 5.1.9

A function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ defined on an open set U containing $\mathbf{A}$ is **differentiable at $\mathbf{A}$** if

$$\mathbf{F}(\mathbf{A}+\mathbf{H}) - \mathbf{F}(\mathbf{A})$$

can be well approximated by a linear function $\mathbf{L}_\mathbf{A}$ in the sense that

$$\frac{\|\mathbf{F}(\mathbf{A}+\mathbf{H}) - \mathbf{F}(\mathbf{A}) - \mathbf{L}_\mathbf{A}(\mathbf{H})\|}{\|\mathbf{H}\|}$$

tends to zero as $\|\mathbf{H}\| \to 0$.

By an argument similar to the one we gave for $f : \mathbb{R}^2 \to \mathbb{R}$ we can show that:

- If $\mathbf{F}$ is differentiable at $\mathbf{A}$, then $\mathbf{F}$ is continuous at $\mathbf{A}$.
- The component functions f_i have partial derivatives, and the linear function $\mathbf{L}_\mathbf{A}$ can be expressed as:

  $$\mathbf{L}_\mathbf{A}(\mathbf{H}) = D\mathbf{F}(\mathbf{A})\mathbf{H},$$

  where $D\mathbf{F}(\mathbf{A})$ is the matrix (5.8) of partial derivatives.

Example 5.1.10

Let:

$$f(x, y, z) = x^2 \sin(yz)$$

The partial derivatives of f are

$$f_x = 2x \sin(yz), \quad f_y = zx^2 \cos(yz), \quad f_z = yx^2 \cos(yz)$$

At the point $\left(1, \frac{\pi}{2}, 2\right)$, we have:

$$f_x\left(1, \frac{\pi}{2}, 2\right) = 0, \quad f_y\left(1, \frac{\pi}{2}, 2\right) = -2, \quad f_z\left(1, \frac{\pi}{2}, 2\right) = -\frac{\pi}{2}$$

Therefore, the derivative matrix is:

$$Df\left(1, \frac{\pi}{2}, 2\right) = \begin{bmatrix} 0 & -2 & -\frac{\pi}{2} \end{bmatrix}$$

Example 5.1.11

Let:

$$\mathbf{F}(x, y) = \left(x^2 + y^2, x, -y^3\right)$$

Find $D\mathbf{F}(1, -2)$.

$$\frac{\partial f_1}{\partial x} = 2x, \quad \frac{\partial f_1}{\partial y} = 2y$$

$$\frac{\partial f_2}{\partial x} = 1, \quad \frac{\partial f_2}{\partial y} = 0$$

$$\frac{\partial f_3}{\partial x} = 0, \quad \frac{\partial f_3}{\partial y} = -3y^2.$$

At the point $(1, -2)$, evaluate the partial derivatives:

$$D\mathbf{F}(1, -2) = \begin{bmatrix} 2 & -4 \\ 1 & 0 \\ 0 & -12 \end{bmatrix}$$

Definition 5.1.12

For $n = m$ the matrix derivative (5.8) of $\mathbf{F}$ at $\mathbf{A}$ is a square matrix.

Its determinant is called the Jacobian of $\mathbf{F}$ at $\mathbf{A}$ and is denoted as $J\mathbf{F}(\mathbf{A})$:

$$J\mathbf{F}(\mathbf{A}) = \det D\mathbf{F}(\mathbf{A}) \tag{5.9}$$

- Recall the derivative $f'(a)$ can be thought of as a **local stretching factor** by f, since $\frac{f(x)-f(a)}{x-a} \to f'(a)$ as $(x-a) \to 0$.

  $\Downarrow$

  The geometric meaning of the Jacobian is **local magnification** of volume by $\mathbf{F}$.

- Let $B_r(\mathbf{A})$ be the ball of radius r centered at $\mathbf{A}$, and $C_r(\mathbf{A})$ its image under the mapping $\mathbf{F}$.

- The ratio of volumes:

  $$\frac{\mathrm{Vol}(C_r(\mathbf{A}))}{\mathrm{Vol}(B_r(\mathbf{A}))}$$

  tends to $|J\mathbf{F}(\mathbf{A})|$ as $r \to 0$.

Example 5.1.13

Let:

$$\mathbf{F}(x, y) = \left(x^2 + y, y^3 + xy\right)$$

Find the Jacobian of $\mathbf{F}$ at $(1, 2)$, and interpret it as a local magnification of area.

$$D\mathbf{F}(x, y) = \begin{bmatrix} 2x & 1 \\ y & 3y^2 + x \end{bmatrix}, \quad D\mathbf{F}(1, 2) = \begin{bmatrix} 2 & 1 \\ 2 & 13 \end{bmatrix}$$

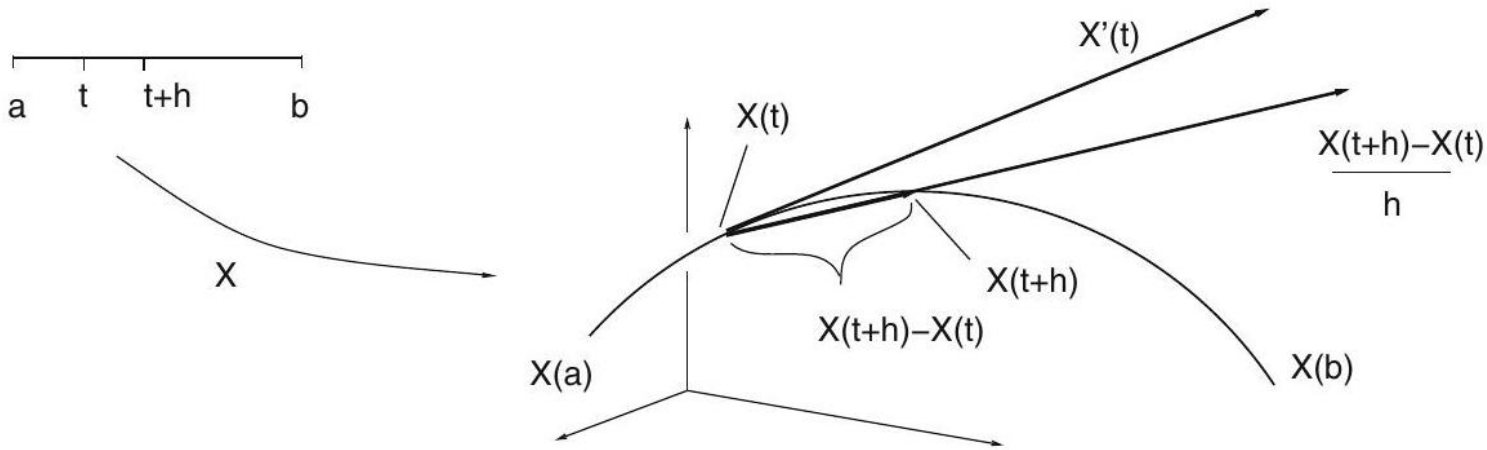
$$J\mathbf{F}(1, 2) = \det \begin{bmatrix} 2 & 1 \\ 2 & 13 \end{bmatrix} = 24$$

The area of the image $\mathbf{F}(B_r)$ of a small disk B_r of radius r centered at $(1, 2)$ is about 24 times as large as the area of B_r.

Derivative of a Vector-Valued Function

If a function $\mathbf{X} : \mathbb{R} \to \mathbb{R}^n$ is differentiable at t, then:

$$\frac{\mathbf{X}(t+h) - \mathbf{X}(t)}{h} - \mathbf{X}'(t) \to 0 \quad \text{as} \quad h \to 0.$$



- The derivative $\mathbf{X}'(t)$ is the limit of secant vectors divided by h, and it is tangent to the curve at $\mathbf{X}(t)$.
- If $\mathbf{X}(t)$ represents the position of a particle at time t, then:
  - $\mathbf{X}'(t)$ is the velocity of the particle at time t.
  - $\|\mathbf{X}'(t)\|$ is the speed of the particle.

Example 5.1.14

Let

$$\mathbf{X}(t) = (\cos t, \sin t, t)$$

The matrix derivative of $\mathbf{X}$ at t is

$$D\mathbf{X}(t) = \mathbf{X}'(t) = \begin{bmatrix} -\sin t \\ \cos t \\ 1 \end{bmatrix}$$

Thinking of $\mathbf{X}'(t)$ as the velocity vector at time t, we may write

$$\mathbf{X}'(t) = (-\sin t, \cos t, 1)$$

Then, its speed is

$$\|\mathbf{X}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

Definition 5.1.15

The vector of partial derivatives of a function $f : \mathbb{R}^n \to \mathbb{R}$ is denoted

$$\nabla f = (f_{x_1}, \ldots, f_{x_n}) \text{ or } \operatorname{grad} f$$

and is called the gradient of f.

Example 5.1.16

Let

$$f(x, y, z) = x^2 \sin(yz)$$

By the calculations in **Example 5.1.10**,

$$\nabla f(x, y, z) = \left(2x \sin(yz), zx^2 \cos(yz), yx^2 \cos(yz)\right)$$

$$\nabla f\left(1, \frac{\pi}{2}, 2\right) = \left(0, -2, -\frac{\pi}{2}\right)$$

Using the gradient notation, the linear approximation of $f(\mathbf{A} + \mathbf{H})$ at $\mathbf{A}$ can be written using the dot product:

$$f(\mathbf{A} + \mathbf{H}) \approx f(\mathbf{A}) + \nabla f(\mathbf{A}) \cdot \mathbf{H}$$



Definition 5.1.17

A function $\mathbf{F}$ from an open set U of $\mathbb{R}^n$ to $\mathbb{R}^m$ is called **continuously differentiable** if it has partial derivatives that are continuous functions on U.

A continuously differentiable function is called a C^1 function.

↳ Differentiable once $\Rightarrow C^2$ C^k

In **Theorem 5.1.8** we have shown that a function of two variables that has continuous partial derivatives is differentiable.

The analogous theorem holds for functions of several variables.

can take second derivatives

Theorem 5.1.18

If $\mathbf{F}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ is C^1 on an open set U, then $\mathbf{F}$ is differentiable at each point of U.

Section 2

chain Rule.

The tangent plane and partial derivatives

Geometric Interpretation of Partial Derivatives

For a function f of a single variable, the derivative $f'(a)$ is the slope of the tangent line:

$$y = f(a) + f'(a)(x - a).$$

For a function f of two variables, suppose f is differentiable at (a, b), with partial derivatives $f_x(a, b)$ and $f_y(a, b)$ at (a, b).

The geometric interpretation of linear approximation is that the graph of

$$z = L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is the plane tangent to the graph of f at the point $(a, b, f(a, b))$.

Rewriting the equation

$$f_x(a, b)(x - a) + f_y(a, b)(y - b) + (-1)(z - f(a, b)) = 0 \qquad (5.10)$$

Continue

- From equation (5.10), a normal vector to the tangent plane is:

$$\mathbf{N} = (f_x(a,b), f_y(a,b), -1)\,.$$

- If f is differentiable at (a,b), then we say $\mathbf{N}$ is normal to the graph of f at $(a,b,f(a,b))$.

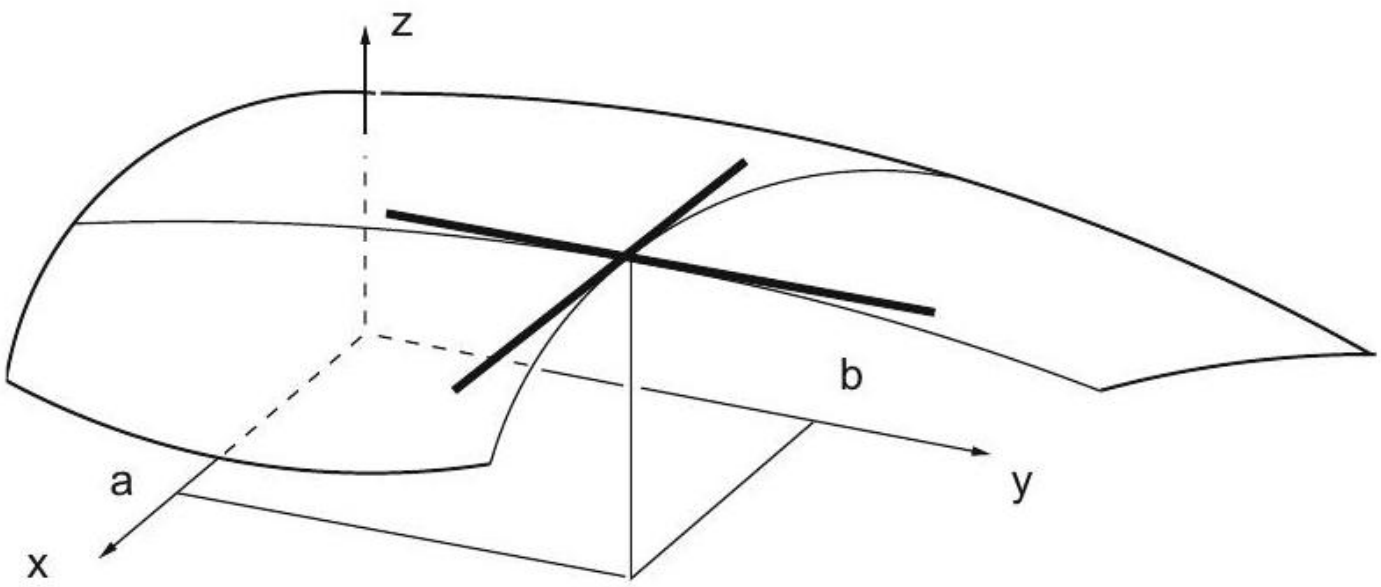


Figure 5.1: A graph partly cut away for clarity at its intersections with planes $x = a$ and $y = b$.

Continue

Another way to obtain an equation for the plane tangent to the graph of f at (a, b) is:

1. Intersect the graph of f with the planes $x = a$ and $y = b$ (see Figure 5.1)
2. Use the lines tangent to the resulting curves to determine the tangent plane

In the plane $x = a$, an equation for the line tangent to the curve at $(a, b, f(a, b))$ is:

$$c_1(t) = (a, b, f(a, b)) + t\,(0, 1, f_y(a, b))$$

In the plane $y = b$, an equation for the line tangent to the intersection curve is:

$$c_2(s) = (a, b, f(a, b)) + s\,(1, 0, f_x(a, b))$$

Parametric Equation and Normal Vector

A parametric equation for the plane determined by these two lines is:

$$\mathbf{P}(s,t) = (a, b, f(a,b)) + s\,(1, 0, f_x(a,b)) + t\,(0, 1, f_y(a,b))$$

A normal vector to this plane is

$$(f_x(a,b), f_y(a,b), -1)\,,$$

since for every $s, t, \tilde{s}, \tilde{t}$:

$$\begin{aligned}\big(\mathbf{P}(\tilde{s},\tilde{t}) - \mathbf{P}(s,t)\big) \cdot (f_x(a,b), f_y(a,b), -1) \\ = (\tilde{s}-s)f_x(a,b) + (\tilde{s}-s)(-f_x(a,b)) \\ + (\tilde{t}-t)f_y(a,b) + (\tilde{t}-t)(-f_y(a,b)) = 0\end{aligned}$$

Example 5.2.1

Let:

$$f(x, y) = x^{1/2} y^{1/3}$$

(a) Find an equation of the plane tangent to the graph of f at $(1, -1)$.

(b) Use the linear approximation of f at $(1, -1)$ to estimate $f(0.9, -1.1)$.

Solution:

We have

$$f_y(x,y) = \frac{1}{3}x^{1/2}y^{-2/3} \quad \text{and} \quad f_x(x,y) = \frac{1}{2}x^{-1/2}y^{1/3}.$$

Since f_x and f_y are continuous near $(1,-1)$, f is differentiable at $(1,-1)$.

$$f(1,-1) = -1, \quad f_y(1,-1) = \frac{1}{3}, \quad f_x(1,-1) = -\frac{1}{2}$$

Solution (Continued)

(a) An equation for the plane tangent to the graph of f at $(1, -1)$ is

$$\begin{aligned} z &= f(1,-1) + f_x(1,-1)(x-1) + f_y(1,-1)(y-(-1)) \\ &= -1 - \frac{1}{2}(x-1) + \frac{1}{3}(y+1), \end{aligned}$$

which simplifies to:

$$3x - 2y + 6z + 1 = 0.$$

(b) To approximate $f(0.9, -1.1)$, use the linear approximation

$$L(x,y) = f(1,-1) + f_x(1,-1)(x-1) + f_y(1,-1)(y-(-1)).$$

$$L(0.9,-1.1) = -1 - \frac{1}{2}(0.9-1) + \frac{1}{3}(-1.1+1) = -0.9833\ldots,$$

a good approximation of $f(0.9, -1.1) = -0.9793\ldots$.

Example 5.2.2

Find all points on the graph of

$$f(x, y) = x^2 - 2xy - y^2 + 6x - 6y$$

where the tangent plane is horizontal.

1. The function f has continuous partial derivatives

$$f_x = 2x - 2y + 6 \quad \text{and} \quad f_y = -2x - 2y - 6,$$

   so f is differentiable at every point.

2. A tangent plane is horizontal if $f_x(x, y) = 0$ and $f_y(x, y) = 0$.
   The equations

$$2x - 2y + 6 = 0 \quad \text{and} \quad -2x - 2y - 6 = 0$$

   have the solution $y = 0$ and $x = -3$.

3. There is one point on the graph with horizontal tangent plane, $(-3, 0, -9)$.

Example 5.2.2 (Continued)

Remark: If two C^1 functions f and g satisfy

$$f(a,b) = g(a,b) \quad \text{and} \quad \nabla f(a,b) = \nabla g(a,b),$$

then their graphs share the same tangent plane at $(a, b, f(a,b))$, and we say that the graphs are **tangent at that point**.

Example 5.2.3

To find points of tangency and the common tangent plane of the graphs of

$$f(x,y) = x^2 - 2xy - y^2$$
$$g(x,y) = x^2 - 3xy + 4x - 16$$

1. Compute the normal vectors:

$$\mathbf{N}_f = (2x - 2y, -2x - 2y, -1), \quad \mathbf{N}_g = (2x - 3y + 4, -3x, -1).$$

2. Since the third components are equal this only happens when

$$2x - 2y = 2x - 3y + 4 \quad \text{and} \quad -2x - 2y = -3x$$

   which are satisfied by $y = 4$ and $x = 8$.
3. Verify that the point $(8, 4, -16)$ is on both graphs. Therefore there is a common tangent plane

$$z = -16 + 8(x - 8) - 24(y - 4)$$

   at the point $(8, 4, -16)$.

Section 3

The chain rule

Theorem 5.3.1 (Chain Rule 1 (for curves))

Let f from $\mathbb{R}^2$ to $\mathbb{R}$ be continuously differentiable on an open set U, and let

$$\mathbf{X}(t) = (x(t), y(t))$$

be a differentiable function from an open interval I to U.

Then the composition $f(x(t), y(t))$ is a differentiable function from I to $\mathbb{R}$ and

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t), y(t)) = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

The right side can be written as the dot product

$$= \nabla f(x(t), y(t)) \cdot \mathbf{X}'(t)$$

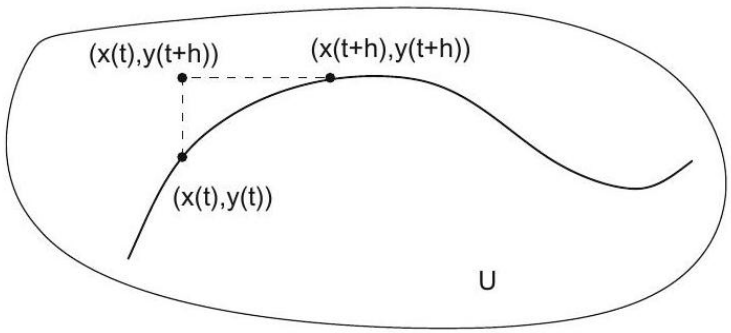


Figure 5.2: Line segments used in the proof of Theorem 5.3.1.

Proof

Since $\mathbf{X}$ is continuous, for sufficiently small $h \neq 0$, the line segments

$$(x(t), y(t)) \to (x(t), y(t+h)) \quad \text{and} \quad (x(t), y(t+h)) \to (x(t+h), y(t+h))$$

remain within U (see Figure 5.2).

1. Rewrite
$$f(x(t+h), y(t+h)) - f(x(t), y(t))$$
as
$$(f(x(t+h), y(t+h)) - f(x(t), y(t+h))) + (f(x(t), y(t+h)) - f(x(t), y(t))).$$

2. By the Mean Value Theorem:
   - There is a number x_* between $x(t+h)$ and $x(t)$ for which
   $$f(x(t+h), y(t+h)) - f(x(t), y(t+h)) = f_x(x_*, y(t+h))(x(t+h) - x(t))$$
   - There is a number y_* between $y(t)$ and $y(t+h)$ for which
   $$f(x(t), y(t+h)) - f(x(t), y(t)) = f_y(x(t), y_*)(y(t+h) - y(t))$$

Proof (Continued)

3. Dividing by h,

$$\frac{f(x(t+h), y(t+h)) - f(x(t), y(t))}{h}$$
$$= \frac{f_x\left(x_*, y(t+h)\right)\left(x(t+h) - x(t)\right)}{h} + \frac{f_y\left(x(t), y_*\right)\left(y(t+h) - y(t)\right)}{h},$$

where $\dfrac{x(t+h) - x(t)}{h} \to x'(t)$ and $\dfrac{y(t+h) - y(t)}{h} \to y'(t)$ as $h \to 0$.

4. Since f_x and f_y are continuous, then as $h \to 0$:

$$f_x\left(x_*, y(t+h)\right) \to f_x(x(t), y(t)) \qquad f_y\left(x(t), y_*\right) \to f_y(x(t), y(t))$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t), y(t)) = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t) = \nabla f(x(t), y(t)) \cdot \mathbf{X}'(t)$$

Example 5.3.2

For $0 \leq t \leq 1$,

$$\mathbf{X}(t) = (x(t), y(t)) = (t, 2t)$$

is the straight path from the origin to point $(1, 2)$.

Let

$$f(x, y) = x^2 + y^4$$

The derivatives of $\mathbf{X}$ and f are $\mathbf{X}'(t) = (1, 2), \nabla f = \left(2x, 4y^3\right)$.

By the Chain Rule,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t)) = \nabla f(x(t), y(t)) \cdot \mathbf{X}'(t) = \left(2(t), 4(2t)^3\right) \cdot (1, 2) = 2t + 64t^3$$

Alternatively, the composite of f and $\mathbf{X}$ is

$$f(\mathbf{X}(t)) = t^2 + 16t^4$$

and the derivative is $\frac{\mathrm{d}}{\mathrm{d}t}\left(t^2 + 16t^4\right) = 2t + 64t^3$ as we found using the Chain Rule.

The analogous theorem can be proved in n dimensions.

Theorem 5.3.3 (Chain Rule 1 (for curves))

Let f from $\mathbb{R}^n$ to $\mathbb{R}$ have continuous partial derivatives on an open set U, and let

$$\mathbf{X}(t) = (x_1(t), \ldots, x_n(t))$$

be a differentiable function from an open interval I in $\mathbb{R}$ into U.

Then the composite $f(\mathbf{X}(t))$ is differentiable on I and

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t)) = \frac{\partial f}{\partial x_1}\frac{\mathrm{d}x_1}{\mathrm{d}t} + \cdots + \frac{\partial f}{\partial x_n}\frac{\mathrm{d}x_n}{\mathrm{d}t} = \nabla f(\mathbf{X}(t)) \cdot \mathbf{X}'(t) = Df(\mathbf{X}(t))D\mathbf{X}(t).$$

Example 5.3.4

For $-\infty < t < \infty$, let

$$\mathbf{X}(t) = (\cos t, \sin t, t)$$

be a curve in $\mathbb{R}^3$ that gives the position of a particle at time t. Let

$$f(x, y, z) = \mathrm{e}^{-x^2-y^2-z^2}$$

represent the temperature of the particle at point (x, y, z).

- The composite

  $$f(\mathbf{X}(t)) = \mathrm{e}^{-\left(1+t^2\right)}$$

  is the temperature of the particle at time t.
- The rate at which the temperature of the particle changes with respect to time is

  $$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t)) = -2t\mathrm{e}^{-\left(1+t^2\right)}$$

Example 5.3.5 (Continued)

- Calculate derivatives: the velocity of the particle at time t,

$$\mathbf{X}'(t) = (-\sin t, \cos t, 1)$$

- The gradient of the temperature at (x, y, z)

$$\nabla f(x, y, z) = -2\mathrm{e}^{-x^2-y^2-z^2}(x, y, z)$$

- The rate at which the temperature of the particle changes with respect to time is

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t)) &= \nabla f(\mathbf{X}(t)) \cdot \mathbf{X}'(t) \\ &= -2\mathrm{e}^{-\left(1+t^2\right)}(\cos t, \sin t, t) \cdot (-\sin t, \cos t, 1) \\ &= -2t\mathrm{e}^{-\left(1+t^2\right)}\end{aligned}$$

Directional derivative

The rate at which a function f from $\mathbb{R}^n$ to $\mathbb{R}$ changes as we move from a point $\mathbf{P}$ along the straight line $\mathbf{X}(t) = \mathbf{P} + t\mathbf{V}, -h \leq t \leq h$ is

$$\lim_{t\to 0} \frac{f(\mathbf{X}(t)) - f(\mathbf{X}(0))}{t - 0} = \left.\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t))\right|_{t=0}$$

By the Chain Rule for curves

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{X}(t)) = \nabla f(\mathbf{X}(t)) \cdot \mathbf{X}'(t)$$

At $t = 0$

$$\nabla f(\mathbf{X}(0)) \cdot \mathbf{X}'(0) = \nabla f(\mathbf{P}) \cdot \mathbf{V}$$

Definition 5.3.6

Let f be a C^1 function from an open set in $\mathbb{R}^n$ containing $\mathbf{P}$ to $\mathbb{R}$, and let $\mathbf{V}$ be a unit vector in $\mathbb{R}^n$. The **directional derivative** of f at $\mathbf{P}$ in the direction of $\mathbf{V}$ is:

$$D_{\mathbf{V}} f(\mathbf{P}) = \nabla f(\mathbf{P}) \cdot \mathbf{V}.$$

- The directional derivative gives insight into the gradient of f at $\mathbf{P}$.
- Since $\|\mathbf{V}\| = 1$, the directional derivative can be written as:

$$D_{\mathbf{V}} f(\mathbf{P}) = \nabla f(\mathbf{P}) \cdot \mathbf{V} = \|\nabla f(\mathbf{P})\| \cos\theta$$

  where θ is the angle between $\nabla f(\mathbf{P})$ and $\mathbf{V}$.
- The directional derivative is greatest when $\cos\theta = 1$, i.e., when $\mathbf{V}$ is in the direction of $\nabla f(\mathbf{P})$.
- The unit vector in the direction of the gradient of f at $\mathbf{P}$ is $\frac{\nabla f(\mathbf{P})}{\|\nabla f(\mathbf{P})\|}$, so the greatest directional rate is

$$\nabla f(\mathbf{P}) \cdot \frac{\nabla f(\mathbf{P})}{\|\nabla f(\mathbf{P})\|} = \|\nabla f(\mathbf{P})\|.$$

Example 5.3.7

Let

$$f(x, y, z) = x + y^2 + z^4$$

Find the direction and magnitude of the greatest directional derivative of f at $(3, 2, 1)$.

Example cont'd

Solution.

$$\nabla f = \left(1, 2y, 4z^3\right), \quad \nabla f(3,2,1) = (1,4,4)$$

- The greatest rate of change in f at $(3,2,1)$ is

$$\|\nabla f(3,2,1)\| = \sqrt{1^2 + 4^2 + 4^2} = \sqrt{33} = 5.744\ldots$$

and this occurs in the direction of $(1,4,4)$.

Example 5.3.8

Let

$$f(x, y, z) = x + y^2 + z^4$$

In which directions is the rate of change of f at $(3, 2, 1)$ zero?

Example cont'd

Solution. From **Example 5.3.7**,

$$\nabla f(3,2,1) = (1,4,4)$$

The directional derivative $D_{\mathbf{V}} f(3,2,1)$ is zero if

$$(1,4,4) \cdot \mathbf{V} = 0.$$

That is, if we move from $(3,2,1)$ in any direction orthogonal to the gradient, the rate of change is zero.

The Chain Rule also holds if the first function in the composition is a function of n variables and the second maps $\mathbb{R}$ to $\mathbb{R}$.

Theorem 5.3.9 (Chain Rule 2)

Suppose $y = f(x_1, \ldots, x_n)$ is continuously differentiable on an open set U in $\mathbb{R}^n$ and $z = g(y)$ is continuously differentiable on an open interval V in $\mathbb{R}$ that contains the range of f.

Then the composite $g(f(x_1, \ldots, x_n))$ is continuously differentiable on U and for each x_i

$$\frac{\partial}{\partial x_i} g(f(x_1, \ldots, x_n)) = \frac{\mathrm{d}g}{\mathrm{d}y} \frac{\partial f}{\partial x_i}, \quad (i = 1, \ldots, n).$$

Therefore

$$D(g \circ f)(\mathbf{X}) = \frac{\mathrm{d}g}{\mathrm{d}y} \nabla f(\mathbf{X})$$

Proof

- Since f and g are continuously differentiable on U and V, their partial derivatives exist and are continuous.
- By the Chain Rule for single-variable functions:

$$\frac{\partial z}{\partial x_i} = \frac{\mathrm{d}z}{\mathrm{d}y}\frac{\partial y}{\partial x_i}.$$

- Since $\frac{\partial z}{\partial x_i}$ is a product of continuous functions, it is continuous on U.

The most general form of the Chain Rule applies to compositions of continuously differentiable functions: $\mathbf{X} : \mathbb{R}^k \to \mathbb{R}^m$, $\mathbf{F} : \mathbb{R}^m \to \mathbb{R}^n$.

Theorem 5.3.10 (Chain Rule)

Let U be an open set in $\mathbb{R}^k$, and let V be an open set in $\mathbb{R}^m$. Suppose that

$$\mathbf{X}(\mathbf{T}) = (x_1(t_1, \ldots, t_k), \ldots, x_m(t_1, \ldots, t_k))$$

is continuously differentiable on U,

$$\mathbf{F}(\mathbf{X}) = (y_1(x_1, \ldots, x_m), \ldots, y_n(x_1, \ldots, x_m))$$

is continuously differentiable on V, and that the range of $\mathbf{X}$ is contained in V.

Then the composite $\mathbf{F}(\mathbf{X}(\mathbf{T}))$ is continuously differentiable on U and the derivative of the composite is the product of the derivative matrices:

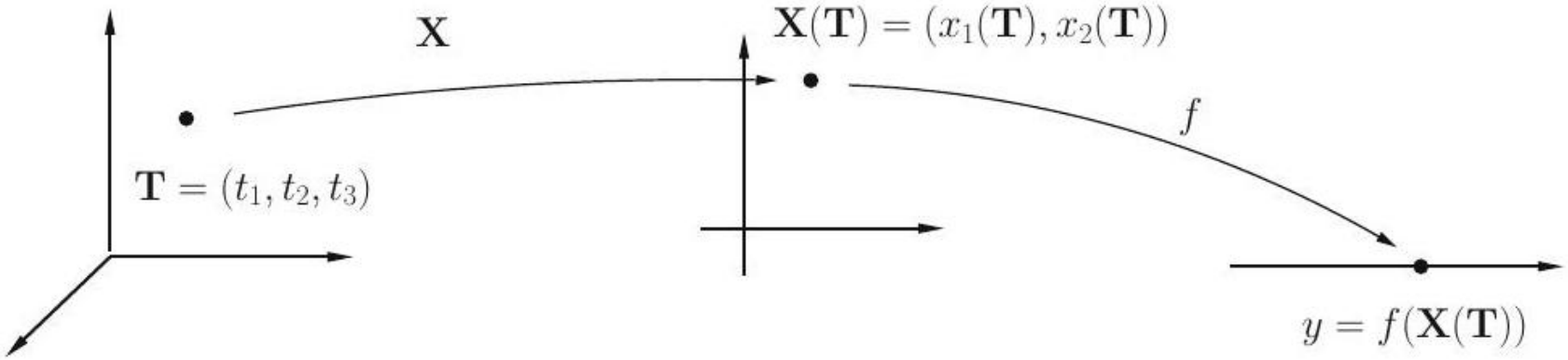
$$D(\mathbf{F} \circ \mathbf{X})(\mathbf{T}) = D\mathbf{F}(\mathbf{X}(\mathbf{T}))D\mathbf{X}(\mathbf{T})$$

Componentwise we can write this relation as

$$\frac{\partial y_i}{\partial t_j} = \frac{\partial y_i}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial y_i}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial y_i}{\partial x_m}\frac{\partial x_m}{\partial t_j}$$

Continue

This figure illustrates the composition of a function from $\mathbb{R}^3$ to $\mathbb{R}^2$ and a function from $\mathbb{R}^2$ to $\mathbb{R}$.



Proof:

- Consider each component function $f_i(\mathbf{X}(t)) = y_i$ as a function of a single variable t_j by holding the other t_k constant.

  Use the Chain Rule for curves, **Theorem 5.2.6**, to find the partial derivative of the composite $f_i(\mathbf{X}(\mathbf{T}))$ with respect to t_j. This gives

$$\frac{\partial y_i}{\partial t_j} = \frac{\partial y_i}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial y_i}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial y_i}{\partial x_m}\frac{\partial x_m}{\partial t_j} \tag{5.12}$$

Proof (Continued)

- Since we have assumed that the partial derivatives of $\mathbf{X}$ and $\mathbf{F}$ are continuous, each such partial derivative is continuous on U.

  Hence $\mathbf{F} \circ \mathbf{X}$ is C^1 on U. By **Theorem 5.1.19**, $\mathbf{F} \circ \mathbf{X}$ is differentiable on $\mathbf{U}$.

- The partial derivative $\frac{\partial y_i}{\partial t_j}$ is by definition the i, j entry of the matrix

$$D(\mathbf{F} \circ \mathbf{X})(\mathbf{T})$$

- According to (5.12) it is also the i, j entry of the product

$$D\mathbf{F}(\mathbf{X}(\mathbf{T}))D\mathbf{X}(\mathbf{T})$$

where

$$D\mathbf{F}(\mathbf{X}) = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \cdots \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \cdots & \cdots & \frac{\partial y_n}{\partial x_m} \end{bmatrix}, \quad D\mathbf{X}(\mathbf{T}) = \begin{bmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} & \cdots & \cdots \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial x_m}{\partial t_1} & \cdots & \cdots & \frac{\partial x_m}{\partial t_k} \end{bmatrix}.$$

Second derivatives

Definition 5.3.11

A function f defined in an open disk in the x, y plane is **twice continuously differentiable** if:

- Its partial derivatives f_x and f_y exist.
- f_x and f_y have continuous partial derivatives.

The partial derivatives of f_x and f_y are denoted as f_{xx}, f_{xy} and f_{yx}, f_{yy}.

They are called the second partial derivatives of f.

A twice continuously differentiable function is called a C^2 function.

The following result is basic:

Theorem 5.3.12

For a twice continuously differentiable function f on an open disk in the x, y plane, the mixed second partial derivatives are equal:

$$f_{xy} = f_{yx}$$

Proof

Let (a, b) be a point in the open disk where f is twice continuously differentiable, and consider the following combination of translates of f in the x and y directions:

1. Define

$$C(h, k) = f(a + h, b + k) - f(a, b + k) - (f(a + h, b) - f(a, b))$$

and let p and q be the functions

$$p(x) = f(x, b + k) - f(x, b), \quad q(y) = f(a + h, y) - f(a, y).$$

Then we can write $C(h, k)$ in two different ways as

$$\begin{aligned} C(h, k) &= p(a + h) - p(a) \\ &= q(b + k) - q(b). \end{aligned}$$

2. Apply the Mean Value Theorem to $p(x)$ and $q(y)$:
   - There exists h_1 between 0 and h such that:

$$p(a + h) - p(a) = hp'(a + h_1) = h\left(f_x(a + h_1, b + k) - f_x(a + h_1, b)\right).$$

Proof (Continued)

- There exists k_1 between 0 and k such that:

$$f_x(a+h_1, b+k) - f_x(a+h_1, b) = k f_{xy}(a+h_1, b+k_1).$$

- Combining these results:

$$C(h,k) = hk f_{xy}(a+h_1, b+k_1).$$

3. Similarly, apply the Mean Value Theorem to $q(y)$:
   - There exists k_2 between 0 and k such that:

$$q(b+k) - q(b) = k q'(b+k_2) = k\left(f_y(a+h, b+k_2) - f_y(a, b+k_2)\right).$$

   - There exists h_2 between 0 and h such that:

$$f_y(a+h, b+k_2) - f_y(a, b+k_2) = h f_{yx}(a+h_2, b+k_2).$$

   - Combining these results:

$$C(h,k) = hk f_{yx}(a+h_2, b+k_2).$$

Proof (Continued)

4. Equate the two expressions for $C(h,k)$:

$$hkf_{xy}(a+h_1, b+k_1) = hkf_{yx}(a+h_2, b+k_2).$$

Since $hk \neq 0$, we have:

$$f_{xy}(a+h_1, b+k_1) = f_{yx}(a+h_2, b+k_2).$$

5. Take the limit as $h, k \to 0$:

- By continuity of f_{xy} and f_{yx}:

$$\lim_{h,k\to 0} f_{xy}(a+h_1, b+k_1) = f_{xy}(a,b),$$

$$\lim_{h,k\to 0} f_{yx}(a+h_2, b+k_2) = f_{yx}(a,b).$$

- Therefore:

$$f_{xy}(a,b) = f_{yx}(a,b).$$